

Experiment No. 8

Write the Python program to implement the Travelling Salesman Problem

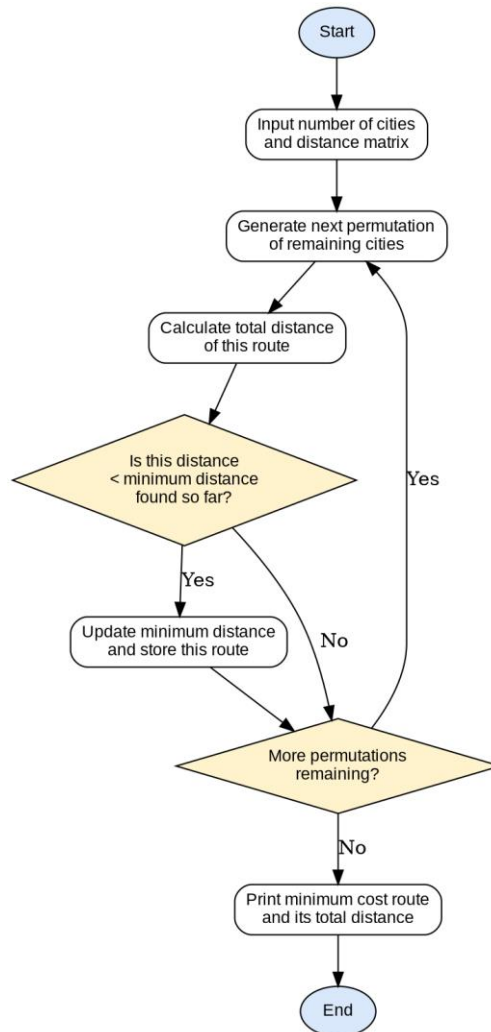
Aim

To write a Python program to solve the Travelling Salesman Problem (TSP) using the brute-force permutation method and find the minimum cost route that visits every city exactly once and returns to the starting city.

Algorithm

1. Start.
2. Input the number of cities and the distance matrix between every pair of cities.
3. Fix the starting city and generate all possible permutations of the remaining cities.
4. For each permutation, form a closed route: start city -> permutation of cities -> start city.
5. Calculate the total distance of the route by summing the distances of consecutive edges.
6. If this total distance is less than the minimum distance found so far, update the minimum distance and store this route as the best route.
7. Repeat steps 4-6 for every remaining permutation.
8. After all permutations are checked, the stored route is the optimal (shortest) route.
9. Display the optimal route and its minimum cost.
10. Stop.

Flowchart



Program

```

from itertools import permutations

def tsp_brute_force(distance_matrix, start=0):
    n = len(distance_matrix)
    cities = list(range(n))
    cities.remove(start)

    min_cost = float('inf')
    best_route = None

    for perm in permutations(cities):
        route = [start] + list(perm) + [start]
        cost = sum(distance_matrix[route[i]][route[i + 1]]
                    for i in range(len(route) - 1))
        if cost < min_cost:
            min_cost = cost
            best_route = route

    return best_route, min_cost

```

```

distance_matrix = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

route, cost = tsp_brute_force(distance_matrix)
print("Optimal Route:", " -> ".join(str(c) for c in route))
print("Minimum Cost:", cost)

```

Result

The program was executed successfully and the output was verified as correct.

Optimal Route: 0 -> 1 -> 3 -> 2 -> 0
 Minimum Cost: 80

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

```

tsp.py
1  from itertools import permutations
2
3  def tsp_brute_force(distance_matrix, start=0):
4      n = len(distance_matrix)
5      cities = list(range(n))
6      cities.remove(start)
7
8      min_cost = float('inf')
9      best_route = None
10
11     for perm in permutations(cities):
12         route = [start] + list(perm) + [start]
13         cost = sum(distance_matrix[route[i]][route[i + 1]]
14                     for i in range(len(route) - 1))
15         if cost < min_cost:
16             min_cost = cost
17             best_route = route
18
19     return best_route, min_cost
20
21
22 distance_matrix = [
23     [0, 10, 15, 20],
24     [10, 0, 35, 25],
25     : (code continues — see Program section)

```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
PS C:/Users/Athavan/AI-Lab> python tsp.py
Optimal Route: 0 -> 1 -> 3 -> 2 -> 0
Minimum Cost: 80
PS C:/Users/Athavan/AI-Lab>

```